IndiGo Airlines

Booking Data Analysis

Revenue Insights · Customer Behaviour · Operational Efficiency

Dataset: 51,201 booking records | 36 variables | 3 seat classes | 2 sales channels

Executive Summary

Key Performance Indicators

₹18.34 Cr

Total Revenue

Booking amount across 50K records

14.89%

Booking Conversion Rate

7,442 completed of 49,964

₹24,521

Avg Booking Amount

Per completed booking out of 7,442

51,201

Total Records

Multi-channel, 3 seat classes

₹4,609

Avg Booking Amt. – Business class

Highest revenue per passenger out of 50K

₹30,995

Avg Booking Amt. – Business class

Highest revenue per passenger out of 7,442

88.1%

Internet Bookings

vs 11% Mobile

73%

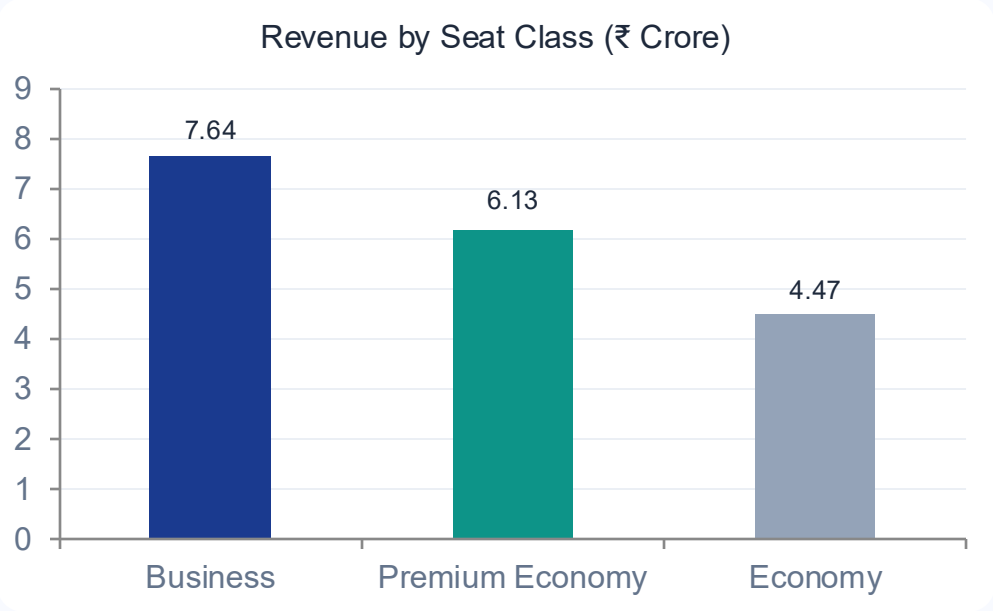
Weekday Passenger Share

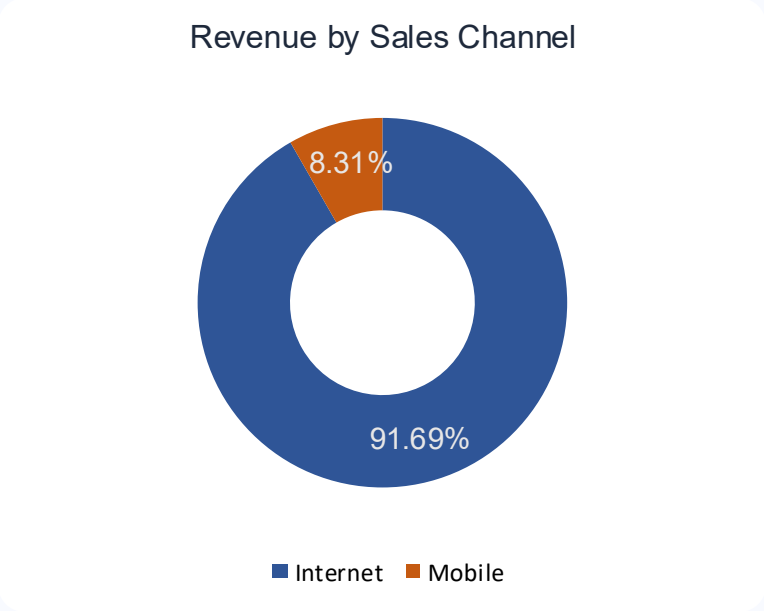
Mon–Thu dominant

This analysis is based on 51,201 booking records across 36 variables, covering three seat classes (Business, Premium Economy, and Economy) and two primary sales channels (Internet and Mobile). The data reveals clear patterns in revenue generation, Customer Behaviour, and operational performance that offer actionable opportunities for IndiGo Airlines.

Revenue Breakdown

Seat Class & Sales Channel





💡 **Key Insight:** Business class generates 41.88% of total revenue despite representing only 33% of passengers. Internet channel dominates with 91.69% of revenue — mobile is underperforming and a major growth opportunity.

Data Types & Metrics Tree

Analytical Framework

Numerical Continuous

flight_duration_planned / actual · booking_amount · purchase_lead · variance_of_flight_time

Numerical Discrete

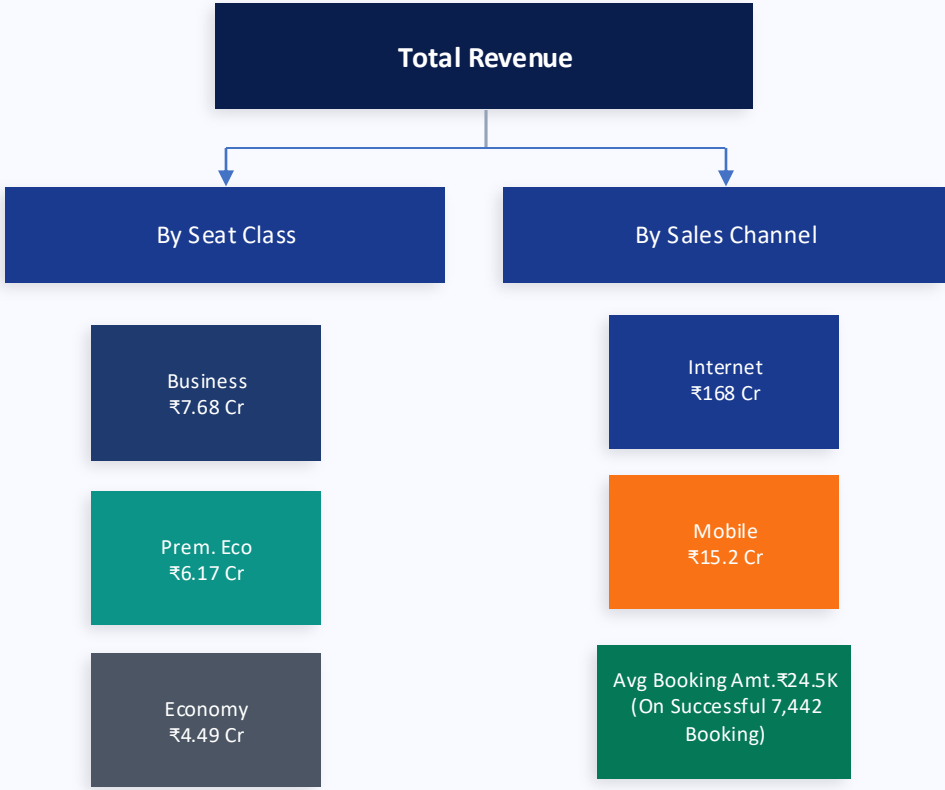
num_passengers · flight_hour · flight_number

Categorical Nominal

sales_channel, Route, seat_class · booking_origin, trip_type

Categorical Ordinal

flight_day (Mon→Sun) · day_segment, length_of_stay_segment



Missing Data Note: 1,214 rows had missing values (2.4%) — handled via exclusion for numeric stats. Sales channel had 59 '{no data}' records removed from channel analysis.

Hypothesis Testing – Customer Preferences

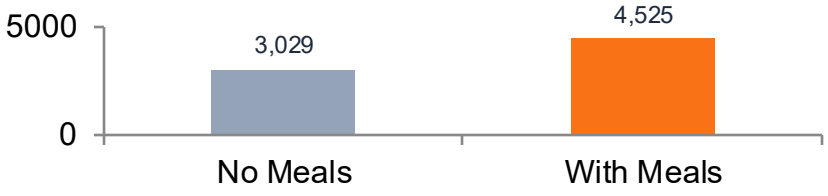
Extra Baggage · In-Flight Meals · Preferred Seat

H1: Extra Baggage → Higher Booking Amount out of 7,442 successful bookings

SUPPORTED BY DATA

H2: In-Flight Meals → Higher Booking Amount

SUPPORTED BY DATA



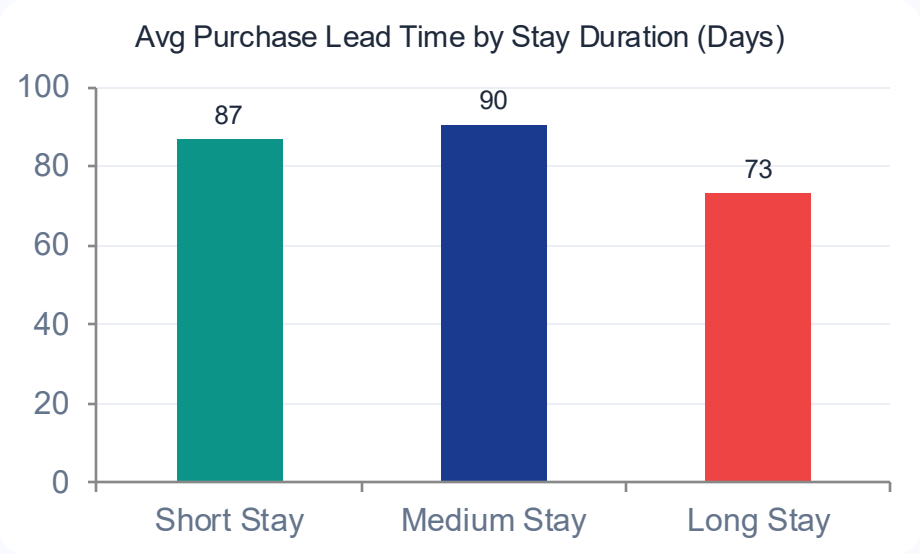
Baggage Conversion Rate by Stay Duration

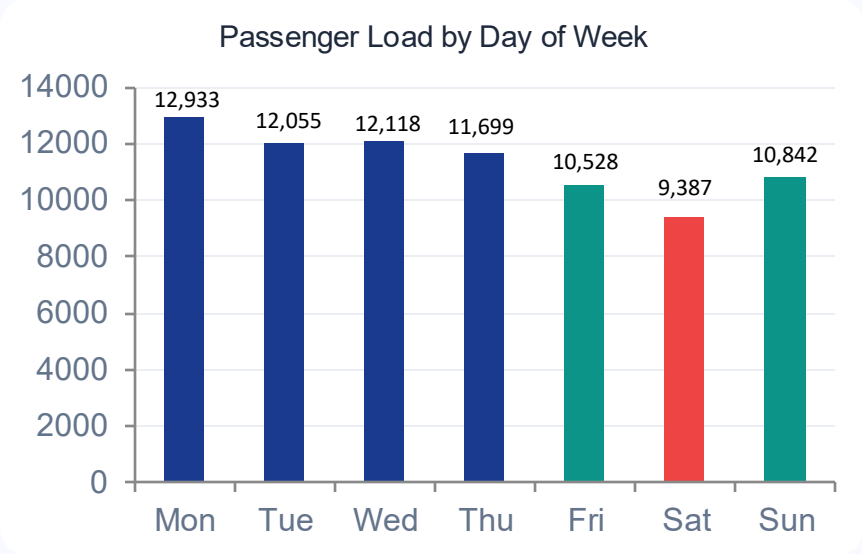
Duration	Booked (baggage)	Total Pax	Conv. Rate	Insight
Stay ≤ 15 days	3,228	13,463	23.98%	Short-stay travellers bundle baggage more
Stay > 15 days	2,321	19,976	11.60%	Longer stays convert at half the rate

Recommendation: Bundle extra baggage incentives into short-stay economy promotions to maximise conversion uplift.

Booking Trends & Purchase Lead Time

When Do Customers Book?





❌ H1 NOT SUPPORTED: Long-stay Passengers book LATER (73 days lead) vs. short-stay (87 days). Short-stay travellers secure seats in advance.

✅ Weekdays dominate: Mon (12,933) vs. Sat (9,387). Saturday is the lowest-load day – pricing opportunity.

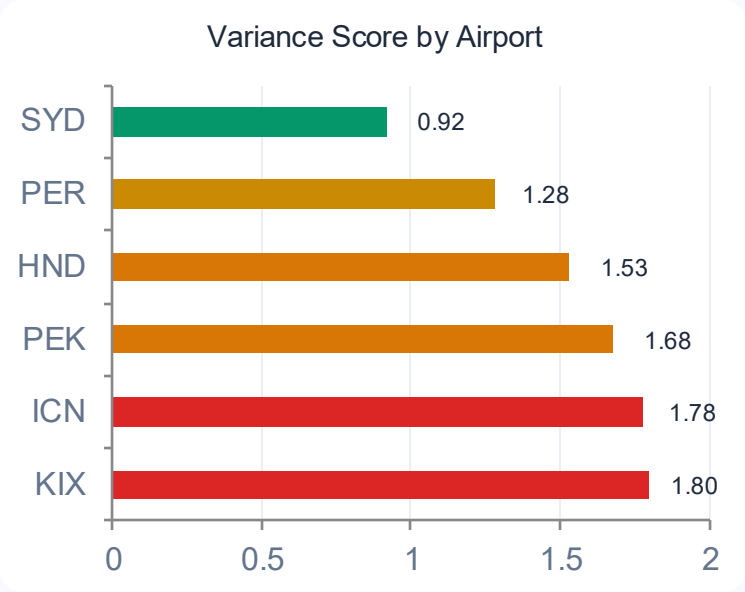
Recommendation: Deploy targeted early-bird promotions for short-stay travellers. Incentivise weekend travel with dynamic pricing discounts on Saturdays.

Operational Efficiency Analysis

Flight Duration Variance – Top Routes

Top 6 Airports with Operational Variance Issues

Rank	Airport	Total Flights	Variance	Status
1	KIX (Kansai)	2,135	1.80	Critical
2	ICN (Incheon)	3,714	1.78	Critical
3	PEK (Beijing)	598	1.68	High
4	HND (Haneda)	2,027	1.53	High
5	PER (Perth)	3,105	1.28	Medium
6	SYD (Sydney)	4,571	0.92	Monitor

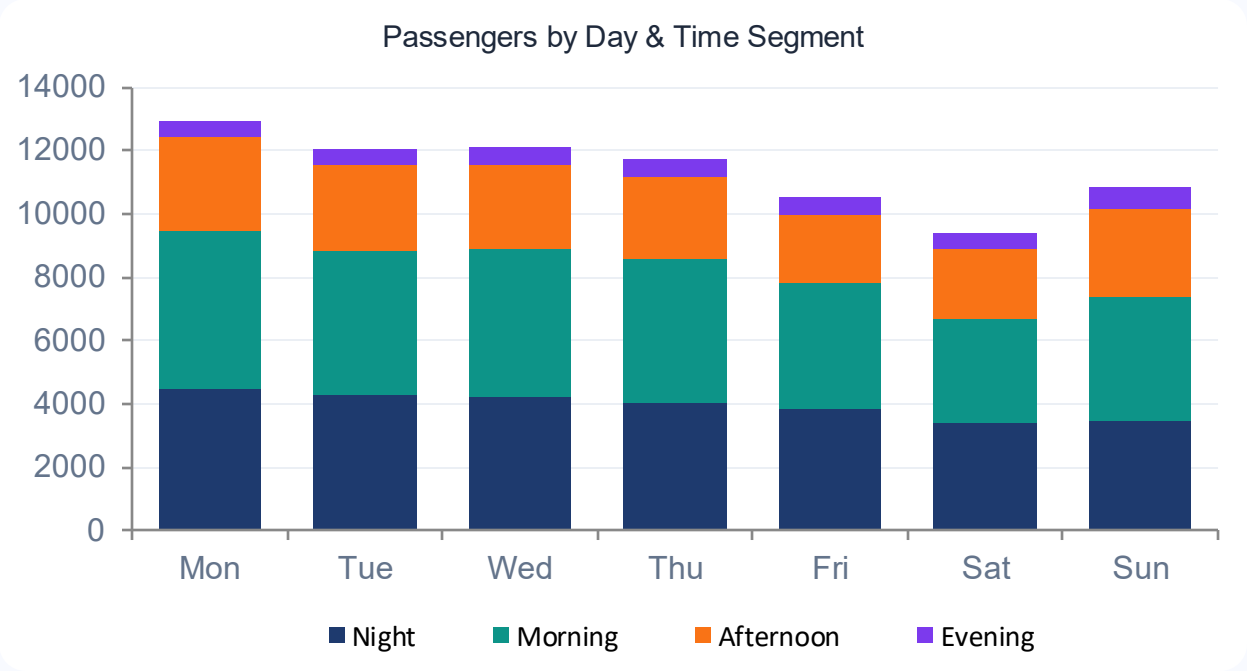


#1 Busiest Route: AKLKUL – 2,680 flights | PENTPE – 924 flights | MELSGN – 842 flights

Recommendation: Escalate KIX and ICN to operational management. Variance reduction reduces fuel & operating costs and improves passenger satisfaction scores.

Passenger Load by Day & Time Segment

When Do Passengers Fly?



Monday

12,933

Peak Day

Saturday

9,387

Lowest Day

Morning (Day_Segment)

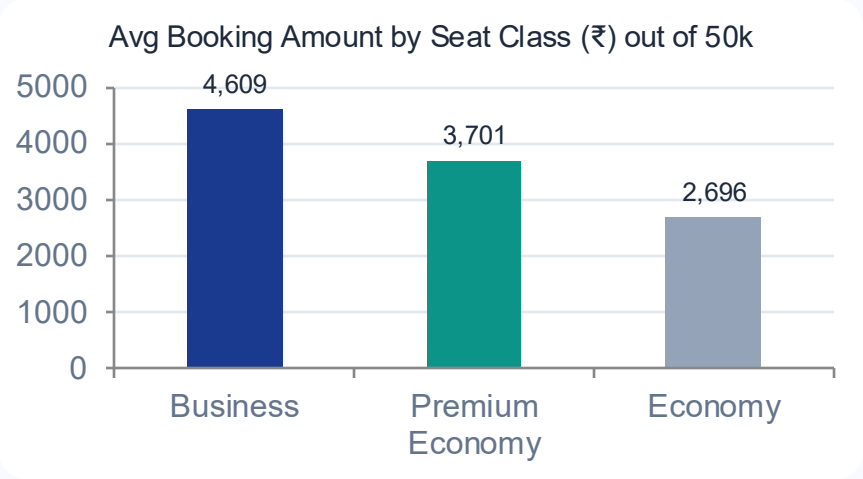
Highest Load Slot

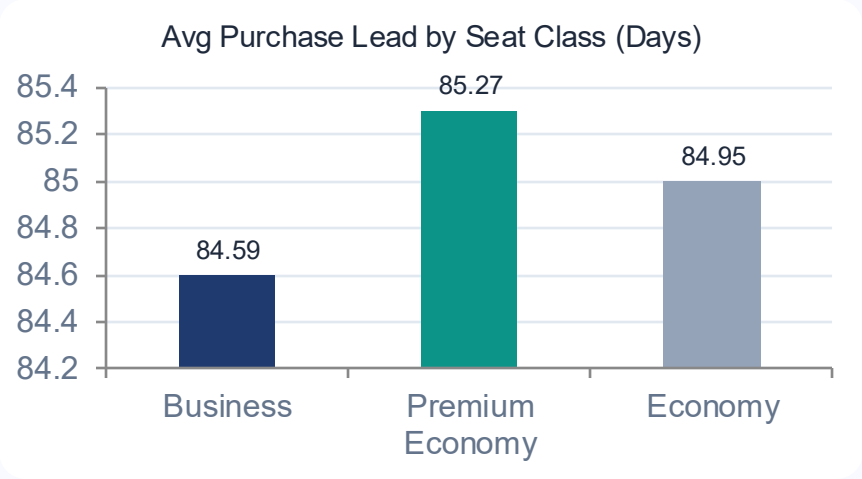
All Weekdays

✅ **H2 SUPPORTED: Morning flights across all weekdays carry the highest passenger load. Weekdays dominate vs. weekends.**

Recommendation: Optimise morning slot pricing on Monday–Thursday with premium pricing. Stimulate Saturday bookings with seat sales & promotions.

Seat Class Deep Dive





✗ **H (Business class books earlier) NOT SUPPORTED:** Purchase lead is nearly identical across all three seat classes (Business: 84.6 days, Premium Economy: 85.3 days, Economy: 85.0 days). Seat class does NOT predict advance booking Behaviour — other factors such as trip purpose or route drive lead time.

Recommendation: Do not segment advance-booking campaigns by seat class. Instead, focus on route type and trip duration as predictors of purchase lead time.

Key Recommendations

Strategic actions derived from booking data analysis

💼 **01**

Monetise Business Class

Business class drives 42% revenue. Introduce loyalty perks, lounge tie-ins & upgrade pathways to grow this segment further.

📱 **02**

Invest in Mobile Channel

Mobile accounts for only 8.3% of revenue. A seamless mobile app experience & app-exclusive deals could double mobile conversions.

🧳 **03**

Bundle Extra Baggage Offers

Passengers booking baggage spend 73% more. Short-stay customers convert at 2× the rate — target them with bundled offers at checkout.

🛫 **04**

Fix Operational Variance

KIX and ICN airports have critical flight duration variance (1.80). Escalate to ops management — reduces fuel cost & improves NPS.

📅 **05**

Dynamic Weekend Pricing

Saturday has 27% fewer passengers than Monday. Stimulate demand with targeted weekend flash sales and early-bird seat promotions.

🌅 **06**

Premium-Price Morning Flights

Morning slots carry the highest passenger load across all weekdays. Introduce peak-time pricing to capture additional revenue per seat.

IndiGo Airlines

Analysis Summary

₹18.34 Cr

Total Revenue Analysed

14.89%

Overall Conversion Rate

6 Airports

Flagged for Ops Review

The data clearly shows that Business class, Internet channel, and morning weekday flights are the primary revenue drivers. IndiGo should prioritise operational fixes at KIX/ICN airports, invest in the mobile channel, and deploy targeted ancillary upsells to maximise per-passenger revenue.

Thank You !
Nishant Kumar Mishra